Neurons That Fire Together Wire Together

# SMARTfit™

*Seize The Now!*

SMARTfit technology builds/repairs neurons
while training the brain and body

# SMARTfit Vision, Reaction and Cognitive Endurance Training for Athletes

Break Through Boundaries with Workouts that put You in the Flow

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# SMARTfit Sports and Tactical Training

- Training of high performers focused on improving vision, speed, reaction and cognitive decision making to increase mental and physical efficiency under pressure.
- Develop vital underlying reaction and motor skills so they are performed perfectly without thought.
- Find the flow quickly and maintain it longer.
- Develop efficient on-the-spot mental processing under pressure especially when physically and mentally exhausted.
- Having the cognitive edge on the competition can mean the difference between winning & losing!

# SMARTfit – Improve Performance

- Process Information Faster
- Substantially Improve Reaction Speed
- Recover Quickly from Mistakes
- Make Instant Decisions Under Pressure
- Increase Attention Span
- Manage Stress Better
- Improve Executive Strategy & Technique Under Pressure

***The Brain Interacts with the World Through the Body***

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# SMARTfit – Comments Made By Elite Athletes

- Improved Cognitive Speed Helps Slow Down the Game
- Improved Timing
- Faster Reaction Skills
- Better Ball Control
- Better Accuracy
- Improved Visual Acuity
- Core Balance Improved Dramatically

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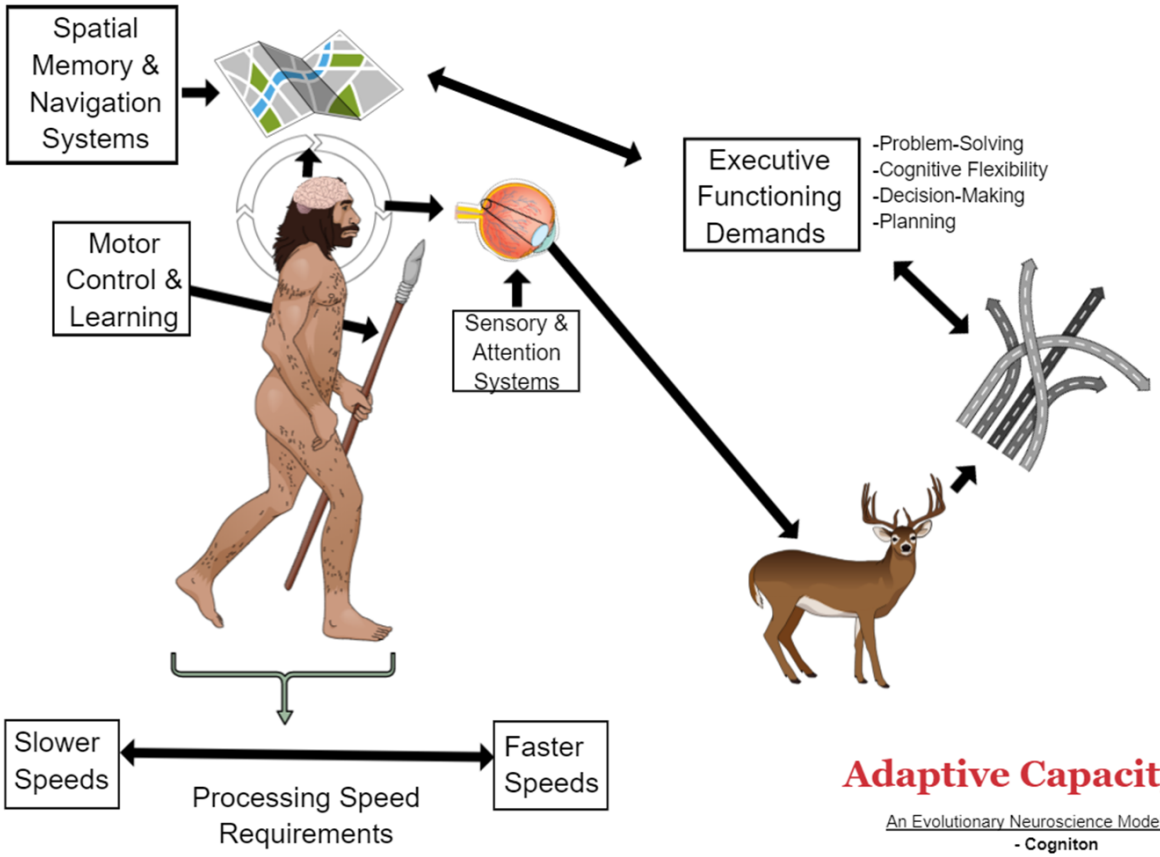
# Goal Centered Training – Aligned With How We Evolved




## Adaptive Capacity Model

An Evolutionary Neuroscience Model Linking

- **Cogniton**
- **Exercise**
- **Brain Health**

*Raichlen et al., 2017, Trends in Neuroscience*

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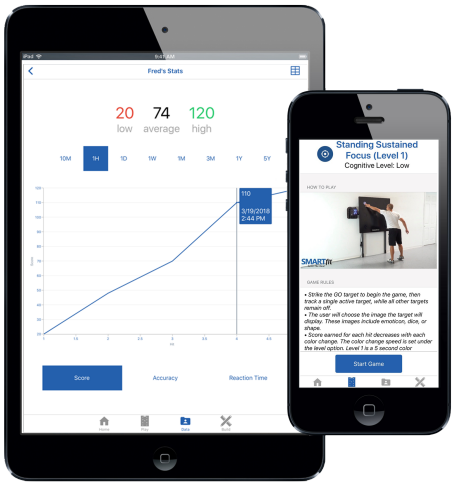
# SMARTfit – Powerful Yet Easy to Customize

Proprietary Operating System

High Quality Hardware



iOS/Android App User Interface

Customized Content

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# What Makes SMARTfit Different?

- Four components of our operating system

**1. The SMARTfit Controller:** capable of processing speed sufficient to match and exceed the fastest human motor performance.

**2. Targets :** Touch sensitive target with display provides visual cues with which the player interacts by performing physical, motor and cognitive tasks. Each target is driven by its own computer.

**3. Sensing System:** Capable of adjusting to tactile range of light to heavy contact used for training proprioception and measuring the visual-cognitive-motor responses. It also trains individuals to manage and apply the appropriate use of power at contact and is capable of identifying accuracy, placement and ball control.

**4. User Interface:** iOS/Android user-friendly interface - set up clients, access 600 video supported turnkey programs, design personalized programs, run the equipment, measure progress and generate baseline and progress reports.

# SMARTfit Performance Video

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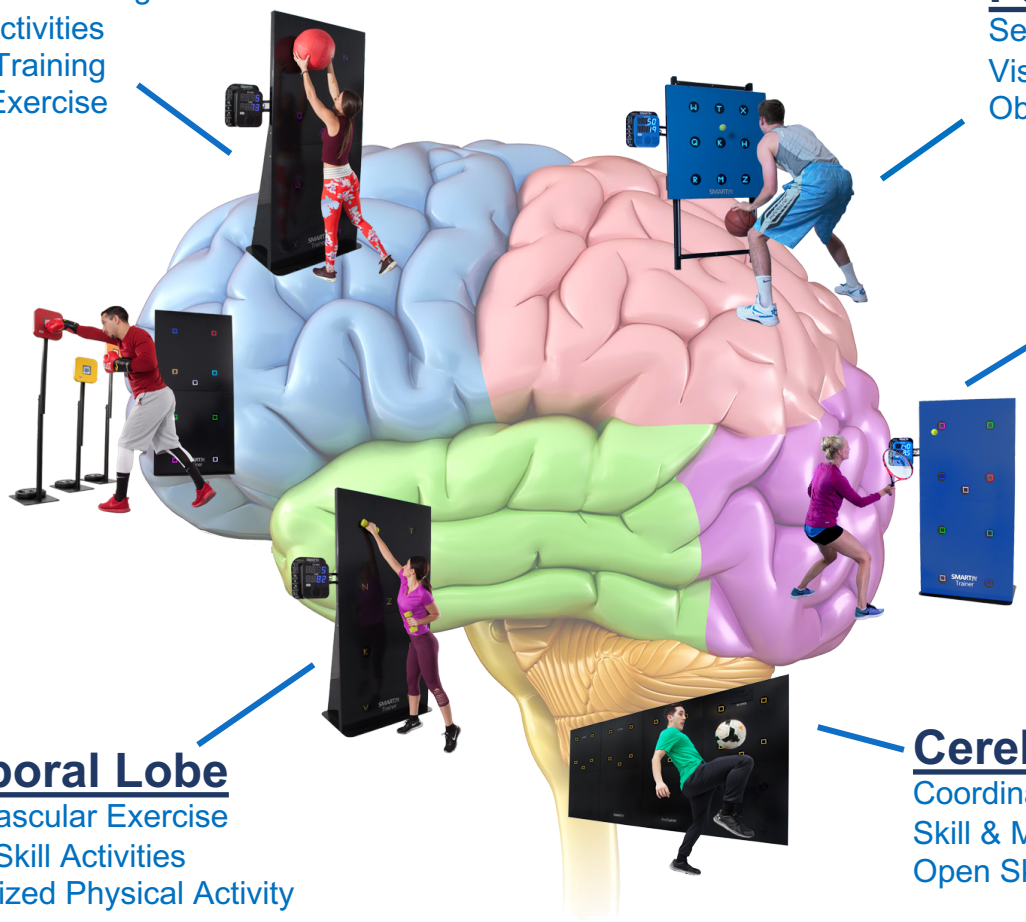
# Efficient – It Needs to Address All Parts of the Brain

## Frontal Lobe
Cognitively Demanding Activities
Open Skill Activities
Resistance Training
Mind-Body Exercise

## Parietal Lobe
Sensory – Rich Activities
Visual – Spatial Demands
Object – Based Activities



## Occipital Lobe
Visual – Spatial Demands
Visual – Attention Demands
Motor Control & Stimulation

## Temporal Lobe
Cardiovascular Exercise
Closed Skill Activities
Generalized Physical Activity

## Cerebellum
Coordinative Exercise
Skill & Motor Learning
Open Skill Activities

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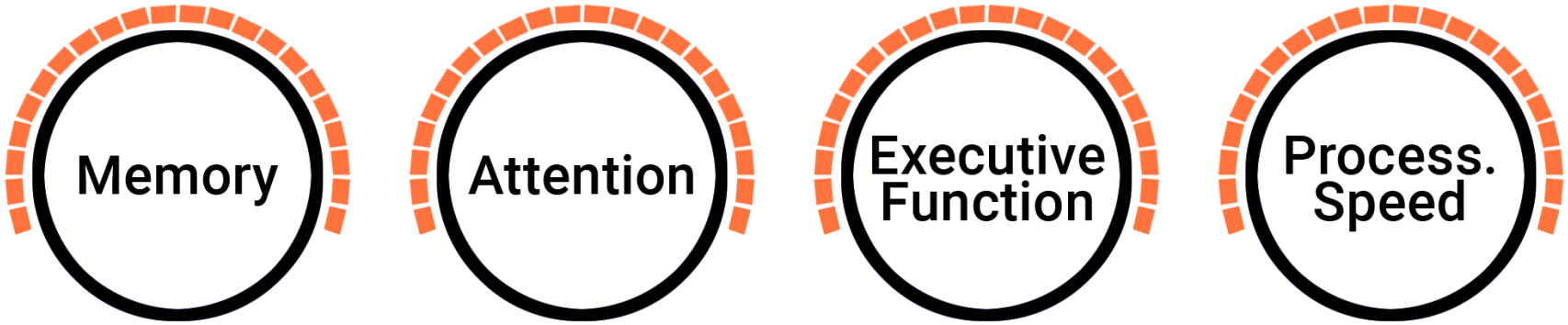
# The Cognitive Domain "Volume Knobs"




"Process Speed" = Processing Speed

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The Volume Knobs are helpful to keep in mind as cognitive-motor techniques are taught

# A Single Session Can Address Key Parts of the Brain

**Memory**

- *Memory Pairs Activities*
- *Pairing Activities*
- *Memory Sequence Activities*

**Attention**

- *Seeking Activities*
- *Tracking Activities*
- *Cognitive Motor Dual-Tasks*

**Processing Speed**

- *Chase the Color*
- *Knock the Lights Out*
- *Tracking Activities*

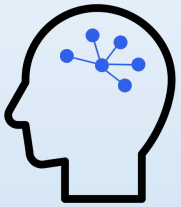
**Executive Functioning**

- *Equations Activities*
- *Track L / R / B*
- *Go No / Go*

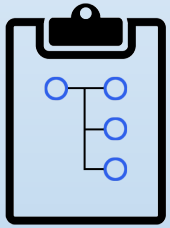
**Language**

- *Spelling*
- *Attn. Letters (Track, Seek)*
- *Memory Letters (Pair, Seq.)*

# Case Study – 5 weeks of training
# 1.5 years of experience on SMARTfit

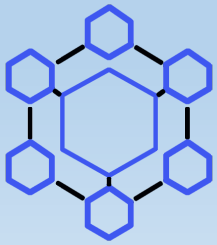


**18%** increase in **Impulse Control**



**11%** increase in **Planning** Skills

**83%** increase in **Executive Function**

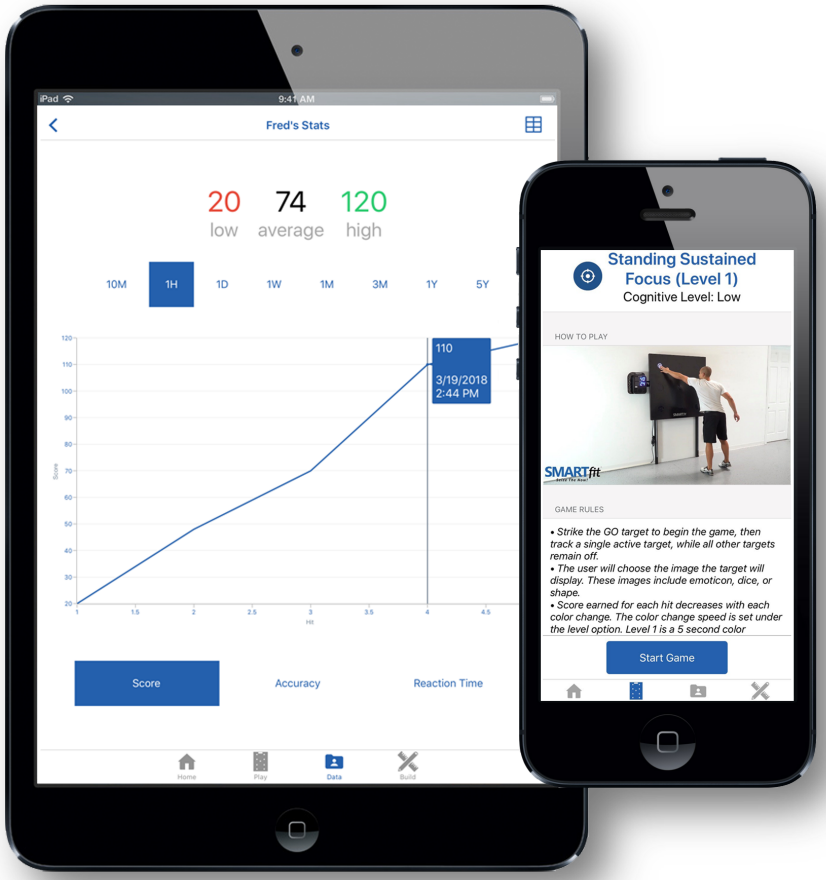


**13%** increase in Visuospatial **Memory**

# The SMARTfit User Interface

## SMARTfit App Key Features

- Trainers Database
- Keep track of athlete stats
- Scoring: High Score, Low Score and Average
- Graph Results
- Comparison over period of time
- Create Categories List of Activities
- Measure Development, Progress & Achievements

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# SMARTfit Professional Systems

SMARTfit ProTrainer

SMARTfit Strike Pods

SMARTfit Mini

SMARTfit Single

SMARTfit Single with Pods

360 degree training

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# SMARTfit's Focus On Psycho-Perceptual-Motor skills

Psychological and expertise skills require:

- specificity,
- progression, and
- overload

over an extended period-of-time in order to be improved. Accordingly, psycho-perceptual-motor skills can deteriorate if not continuously practiced.

This indicates that equal proportions of time should be allocated to the practice of psycho-perceptual-motor skills during weekly training sessions.

# SMARTfit – Taxes Working Memory and Spatial Memory

Specifically designed sports (activities), which tax working memory and spatial ability by incorporating motion in three-dimensional space, are an optimal way to combine the benefits of traditional cognitive training and physical exercise into a single activity.

-Moreau, D., & Conway, A. R. (2013). Cognitive enhancement: a comparative review of computerized and athletic training programs. International Review of Sport and Exercise Psychology, 6(1), 155-183.

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# SMARTfit – Multiple Object Tracking and Dual-task Training

The assessment of MOT (multiple object tracking) in a DT (dual-task) paradigm provided insight in how impaired intelligence constrains the ability of athletes to successfully perform at the highest levels in the complex and dynamical sport-environment.

-Van Biesen, D., Jacobs, L., McCulloch, K., Janssens, L., & Vanlandewijck, Y. C. (2018). Cognitive-motor dual-task ability of athletes with and without intellectual impairment. Journal of sports sciences, 36(5), 513-521.

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# SMARTfit – Dual Task Training

Dynamic dual-task assessments may be more challenging to the postural control system and may better represent postural control performance during dynamic activities.

-Talarico, M. K., Lynall, R. C., Mauntel, T. C., Weinhold, P. S., Padua, D. A., & Mihalik, J. P. (2017). Static and dynamic single leg postural control performance during dual-task paradigms. Journal of sports sciences, 35(11), 1118-1124.

Several studies suggest that the incorporation of a cognitive task into the motor task combines the advantageous effects of cognitive and motor training and leads to greater (motor-) cognitive improvements

- Herold, F., Hamacher, D., Schega, L., & Mueller, N. G. (2018). Thinking while Moving or Moving while Thinking–Concepts of motor-cognitive training for cognitive performance enhancement. Frontiers in aging neuroscience, 10.

# Celebrity Personal Trainer Gunnar Peterson Spokesperson, Program Development and Market Entry Strategist for SMARTfit Inc. and Trainer for the LA Lakers

"SMARTfit merges exercise and performance like nothing else on the market. I hear 'mind/body' all the time in the fitness world, but it wasn't until I saw this technology in action that I realized what mind/body could be for athletes and others of all ages and abilities.

SMARTfit is the real mind/body training tool and I am thrilled to take an active role in programming for this cutting-edge company."

Gunnar Peterson
NSCA- CPT
Duke University Graduate

Gunnar Peterson, Beverly Hills-based personal trainer and custom gym developer whose clients include celebrities and professional athletes from the NBA, NHL, NFL, MLB, USTA, professional boxing and NCAA sports. Peterson serves on the advisory board for *Fitness* magazine, writes for *Muscle and Fitness*, and is regularly featured in numerous high-profile, publications and television shows.

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# SMARTfit - Validated by Kinesiology and Biomechanics

“SMARTfit takes the focus of the player off the mechanics and moves it to the external, forcing the brain to solve two problems at the same time.”

“It solves the big missing part in rehab because we want carry-over onto the field. SMARTfit is a great tool to test if that is happening.”

*Christopher Powers, PhD, PT, FACSM, FAPTA*
*USC Professor*
*Director, Biokinesiology Program*
*Co-Director, Biomechanics Research Lab*

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# SMARTfit Sports Performance Training

**Awaken Brain and Body To Achieve Highly Competitive Physical and Mental Performance Skills**

**Create the Mental & Physical Conditions to Function Efficiently Under Pressure**

**Prevent Injuries**

**Improve Overall Performance**

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